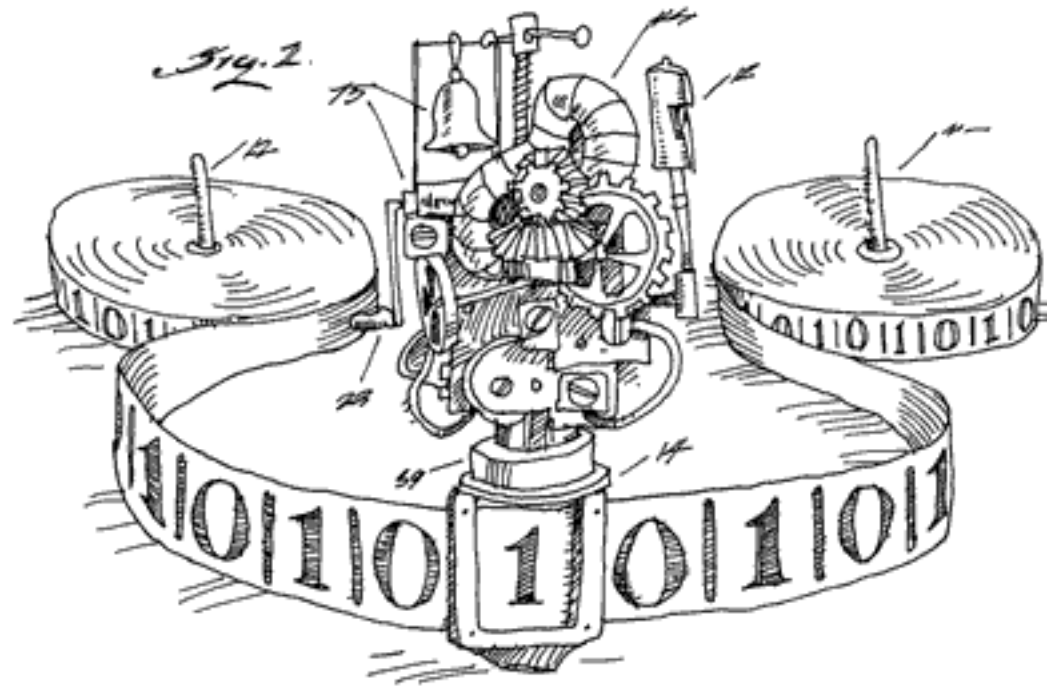


# CSC4010: Introduction to Theoretical Computer Science



## Lecture 8

Adapted from Sipser's MIT OCW Course on TOC

# Last Time

- Is there an explicit computational problem which cannot be solved in any computational problem?
- We showed that  $L_{\text{self}}$ , and  $A_{TM}$  are undecidable via diagonalization.
- A language  $L$  is decidable if and only if  $L$  and  $\bar{L}$  are both recognizable. We used this to conclude that  $\bar{A}_{TM}$  is undecidable.
- We showed  $\text{HALT}_{TM}$  is undecidable by reducing  $A_{TM}$  to  $\text{HALT}_{TM}$ .
- **Proof Strategy:** If there is a decider  $H$  for  $\text{HALT}_{TM}$ , then we can construct a decider  $M'$  for  $A_{TM}$ . But  $A_{TM}$  is undecidable.

# Reducibility Method: Undecidability of $\text{HALT}_{TM}$

- Let  $\text{HALT}_{TM} = \{ \langle M, w \rangle \mid M \text{ halts on } w \}$ .

**Theorem:**  $\text{HALT}_{TM}$  is undecidable.

- **Proof:** Proof by contradiction. Assume there exists  $H$  deciding  $\text{HALT}_{TM}$ .

# Reducibility Method: Undecidability of $\text{HALT}_{TM}$

- Let  $\text{HALT}_{TM} = \{ \langle M, w \rangle \mid M \text{ halts on } w \}$ .

**Theorem:**  $\text{HALT}_{TM}$  is undecidable.

- **Proof:** Proof by contradiction. Assume there exists  $H$  deciding  $\text{HALT}_{TM}$ .

**Idea:** Use  $H$  to construct a Turing Machine  $M'$  deciding  $A_{TM}$ .

# Reducibility Method: Undecidability of $\text{HALT}_{TM}$

- Let  $\text{HALT}_{TM} = \{ \langle M, w \rangle \mid M \text{ halts on } w \}$ .

**Theorem:**  $\text{HALT}_{TM}$  is undecidable.

- Proof:** Proof by contradiction. Assume there exists  $H$  deciding  $\text{HALT}_{TM}$ .

**Idea:** Use  $H$  to construct a Turing Machine  $M'$  deciding  $A_{TM}$ .

$M' =$  “On input  $\langle M, w \rangle$

1. Simulate  $H$  on input  $\langle M, w \rangle$ .
2. Reject if  $H$  rejects.
3. If  $H$  accepts, run  $M$  on  $w$  until it halts.
  - (a) If  $M$  accepts, then accept.
  - (b) If  $M$  rejects, then reject.”

# Reducibility Method: Undecidability of $\text{HALT}_{TM}$

- Let  $\text{HALT}_{TM} = \{ \langle M, w \rangle \mid M \text{ halts on } w \}$ .

**Theorem:**  $\text{HALT}_{TM}$  is undecidable.

- Proof:** Proof by contradiction. Assume there exists  $H$  deciding  $\text{HALT}_{TM}$ .

**Idea:** Use  $H$  to construct a Turing Machine  $M'$  deciding  $A_{TM}$ .

$M' =$  “On input  $\langle M, w \rangle$

1. Simulate  $H$  on input  $\langle M, w \rangle$ .
2. Reject if  $H$  rejects.
3. If  $H$  accepts, run  $M$  on  $w$  until it halts.
  - (a) If  $M$  accepts, then accept.
  - (b) If  $M$  rejects, then reject.”

**Claim:** If  $H$  exists, then  $M'$  decides  $A_{TM}$ .

# Reducibility - Concept

- If we have two languages (or problems)  $A$  and  $B$ , then  $A$  being reducible to  $B$  implies we can use  $B$  to solve  $A$ .
- **Example:** Measuring the area of a rectangle is reducible to measuring the lengths of its sides.
- If  $A$  is reducible to  $B$ , then solving  $B$  gives a solution to  $A$ .
  - Then,  $B$  is easy implies  $A$  is easy.
  - Then,  $A$  is hard implies  $B$  is hard.

**This is the form we will use.**

# Another Example: $\text{EMPTY}_{TM}$

- Let  $\text{EMPTY}_{TM} = \{ \langle M \rangle \mid L(M) = \emptyset \}$ .

**Theorem:**  $\text{EMPTY}_{TM}$  is undecidable.

- **Proof:** Proof by contradiction. Assume there exists  $R$  deciding  $\text{EMPTY}_{TM}$ .

# Another Example: $\text{EMPTY}_{TM}$

- Let  $\text{EMPTY}_{TM} = \{ \langle M \rangle \mid L(M) = \emptyset \}$ .

**Theorem:**  $\text{EMPTY}_{TM}$  is undecidable.

- Proof:** Proof by contradiction. Assume there exists  $R$  deciding  $\text{EMPTY}_{TM}$ .

**Idea:** Use  $R$  to construct a Turing Machine  $M'$  deciding  $A_{TM}$ .

# Another Example: $\text{EMPTY}_{TM}$

- Let  $\text{EMPTY}_{TM} = \{ \langle M \rangle \mid L(M) = \emptyset \}$ .

**Theorem:**  $\text{EMPTY}_{TM}$  is undecidable.

- Proof:** Proof by contradiction. Assume there exists  $R$  deciding  $\text{EMPTY}_{TM}$ .

**Idea:** Use  $R$  to construct a Turing Machine  $M'$  deciding  $A_{TM}$ .

**Attempt 1:** Given input  $\langle M, w \rangle$  for  $M'$ , we pass  $\langle M \rangle$  through  $R$ .

# Another Example: $\text{EMPTY}_{TM}$

- Let  $\text{EMPTY}_{TM} = \{ \langle M \rangle \mid L(M) = \emptyset \}$ .

**Theorem:**  $\text{EMPTY}_{TM}$  is undecidable.

- Proof:** Proof by contradiction. Assume there exists  $R$  deciding  $\text{EMPTY}_{TM}$ .

$$L(M_w) = \begin{cases} \{w\} & \text{if } M \text{ accepts } w \\ \emptyset & \text{if } M \text{ does not accept } w \end{cases}$$

# Another Example: $\text{EMPTY}_{TM}$

- Let  $\text{EMPTY}_{TM} = \{ \langle M \rangle \mid L(M) = \emptyset \}$ .

**Theorem:**  $\text{EMPTY}_{TM}$  is undecidable.

- Proof:** Proof by contradiction. Assume there exists  $R$  deciding  $\text{EMPTY}_{TM}$ .

**Idea:** Use  $R$  to construct a Turing Machine  $M'$  deciding  $A_{TM}$ .

$$L(M_w) = \begin{cases} \{w\} & \text{if } M \text{ accepts } w \\ \emptyset & \text{if } M \text{ does not accept } w \end{cases}$$

# Another Example: $\text{EMPTY}_{TM}$

- Let  $\text{EMPTY}_{TM} = \{ \langle M \rangle \mid L(M) = \emptyset \}$ .

**Theorem:**  $\text{EMPTY}_{TM}$  is undecidable.

- Proof:** Proof by contradiction. Assume there exists  $R$  deciding  $\text{EMPTY}_{TM}$ .

**Idea:** Use  $R$  to construct a Turing Machine  $M'$  deciding  $A_{TM}$ .

**Attempt 2:** Given input  $\langle M, w \rangle$  for  $M'$ , transform  $M$  to  $M_w$  defined as follows:

$$L(M_w) = \begin{cases} \{w\} & \text{if } M \text{ accepts } w \\ \emptyset & \text{if } M \text{ does not accept } w \end{cases}$$

# Another Example: $\text{EMPTY}_{TM}$

- Let  $\text{EMPTY}_{TM} = \{ \langle M \rangle \mid L(M) = \emptyset \}$ .

**Theorem:**  $\text{EMPTY}_{TM}$  is undecidable.

- Proof:** Proof by contradiction. Assume there exists  $R$  deciding  $\text{EMPTY}_{TM}$ .

**Idea:** Use  $R$  to construct a Turing Machine  $M'$  deciding  $A_{TM}$ .

**Attempt 2:** Given input  $\langle M, w \rangle$  for  $M'$ , transform  $M$  to  $M_w$  defined as follows:

$M_w =$  "On input  $x \in \{0,1\}^*$

1. If  $x \neq w$ , then reject.
2. Simulate  $M$  on  $w$ .

$$L(M_w) = \begin{cases} \{w\} & \text{if } M \text{ accepts } w \\ \emptyset & \text{if } M \text{ does not accept } w \end{cases}$$

# Another Example: $\text{EMPTY}_{TM}$

- Let  $\text{EMPTY}_{TM} = \{ \langle M \rangle \mid L(M) = \emptyset \}$ .

**Theorem:**  $\text{EMPTY}_{TM}$  is undecidable.

- Proof:** Proof by contradiction. Assume there exists  $R$  deciding  $\text{EMPTY}_{TM}$ .

$$L(M_w) = \begin{cases} \{w\} & \text{if } M \text{ accepts } w \\ \emptyset & \text{if } M \text{ does not accept } w \end{cases}$$

# Another Example: $\text{EMPTY}_{TM}$

- Let  $\text{EMPTY}_{TM} = \{ \langle M \rangle \mid L(M) = \emptyset \}$ .

**Theorem:**  $\text{EMPTY}_{TM}$  is undecidable.

- Proof:** Proof by contradiction. Assume there exists  $R$  deciding  $\text{EMPTY}_{TM}$ .

**Idea:** Use  $R$  to construct a Turing Machine  $M'$  deciding  $A_{TM}$ .

$$L(M_w) = \begin{cases} \{w\} & \text{if } M \text{ accepts } w \\ \emptyset & \text{if } M \text{ does not accept } w \end{cases}$$

# Another Example: $\text{EMPTY}_{TM}$

- Let  $\text{EMPTY}_{TM} = \{ \langle M \rangle \mid L(M) = \emptyset \}$ .

**Theorem:**  $\text{EMPTY}_{TM}$  is undecidable.

- Proof:** Proof by contradiction. Assume there exists  $R$  deciding  $\text{EMPTY}_{TM}$ .

**Idea:** Use  $R$  to construct a Turing Machine  $M'$  deciding  $A_{TM}$ .

**Attempt 2:** Given input  $\langle M, w \rangle$  for  $M'$ , transform  $M$  to  $M_w$  defined as follows:

$$L(M_w) = \begin{cases} \{w\} & \text{if } M \text{ accepts } w \\ \emptyset & \text{if } M \text{ does not accept } w \end{cases}$$

# Another Example: $\text{EMPTY}_{TM}$

- Let  $\text{EMPTY}_{TM} = \{ \langle M \rangle \mid L(M) = \emptyset \}$ .

**Theorem:**  $\text{EMPTY}_{TM}$  is undecidable.

- Proof:** Proof by contradiction. Assume there exists  $R$  deciding  $\text{EMPTY}_{TM}$ .

**Idea:** Use  $R$  to construct a Turing Machine  $M'$  deciding  $A_{TM}$ .

**Attempt 2:** Given input  $\langle M, w \rangle$  for  $M'$ , transform  $M$  to  $M_w$  defined as follows:

$M' =$  "On input  $\langle M, w \rangle$

1. Transform  $\langle M, w \rangle$  into  $M_w$ .
2. Run  $R$  on  $\langle M_w \rangle$ .
3. **Accept** if  $R$  **rejects**.
4. **Reject** if  $R$  **accepts**.

$$L(M_w) = \begin{cases} \{w\} & \text{if } M \text{ accepts } w \\ \emptyset & \text{if } M \text{ does not accept } w \end{cases}$$

# Another Example: $\text{EMPTY}_{TM}$

- Let  $\text{EMPTY}_{TM} = \{ \langle M \rangle \mid L(M) = \emptyset \}$ .

**Theorem:**  $\text{EMPTY}_{TM}$  is undecidable.

- Proof:** Proof by contradiction. Assume there exists  $R$  deciding  $\text{EMPTY}_{TM}$ .

**Idea:** Use  $R$  to construct a Turing Machine  $M'$  deciding  $A_{TM}$ .

**Attempt 2:** Given input  $\langle M, w \rangle$  for  $M'$ , transform  $M$  to  $M_w$  defined as follows:

$M' =$  "On input  $\langle M, w \rangle$

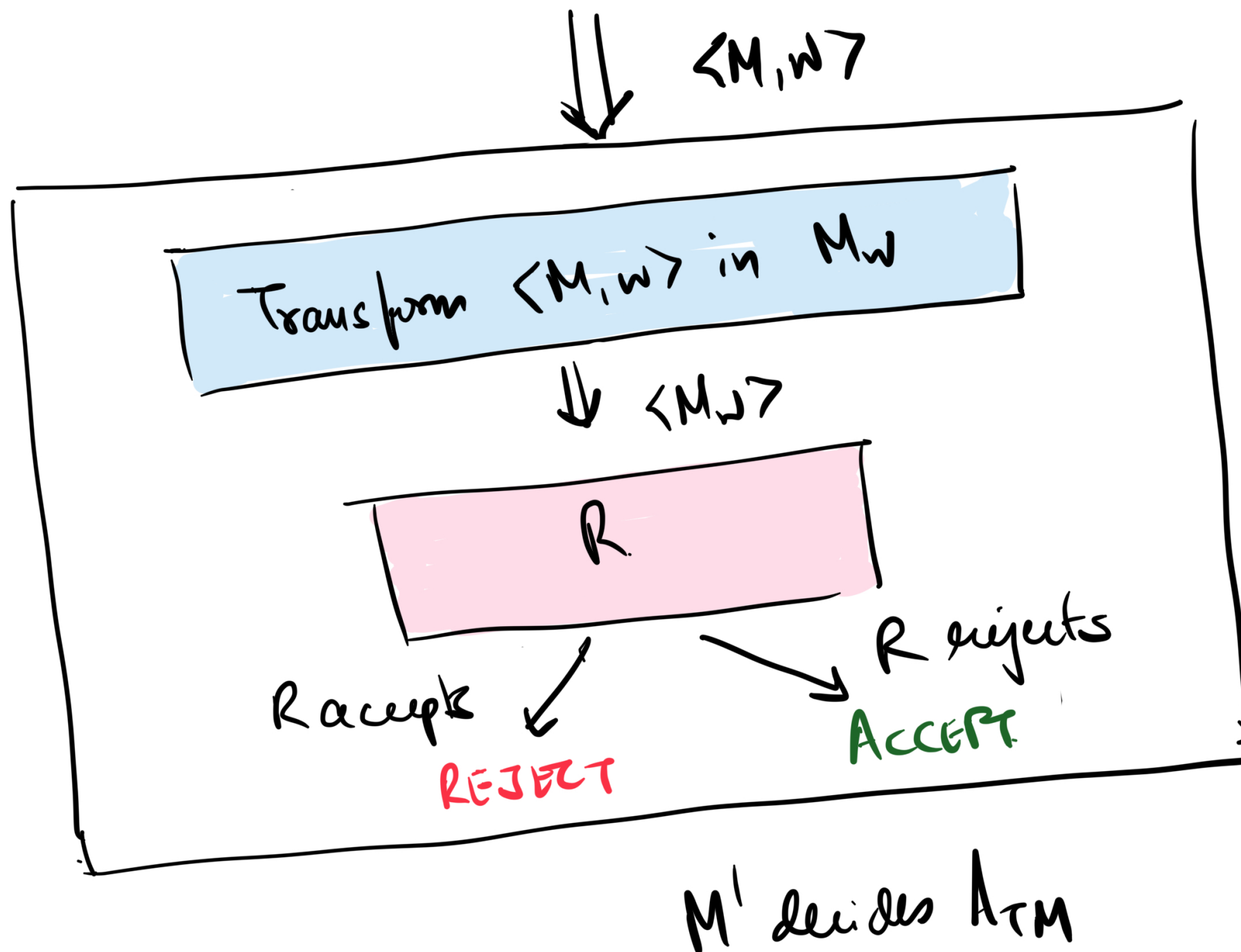
1. Transform  $\langle M, w \rangle$  into  $M_w$ .
2. Run  $R$  on  $\langle M_w \rangle$ .
3. **Accept** if  $R$  **rejects**.
4. **Reject** if  $R$  **accepts**.

$M_w =$  "On input  $x \in \{0,1\}^*$

1. If  $x \neq w$ , then reject.
2. Simulate  $M$  on  $w$ .

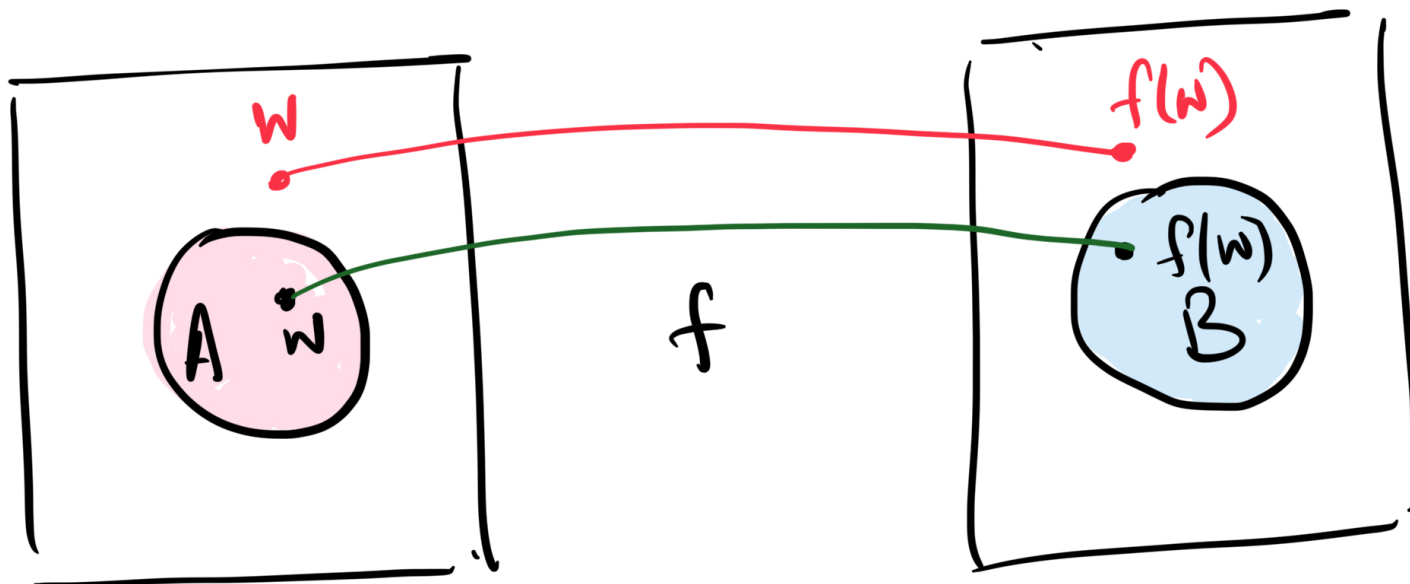
$$L(M_w) = \begin{cases} \{w\} & \text{if } M \text{ accepts } w \\ \emptyset & \text{if } M \text{ does not accept } w \end{cases}$$

# Another Example: $\text{EMPTY}_{\text{TM}}$



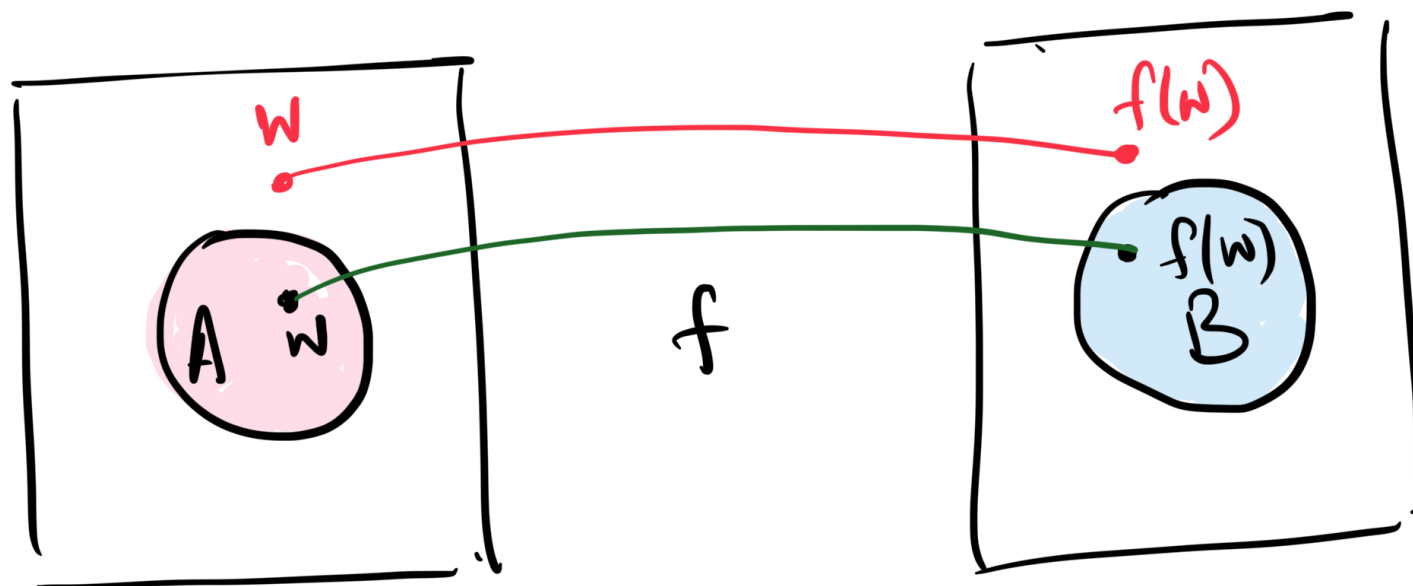
# Mapping Reducibility

- **Definition:** A function  $f : \Sigma^* \rightarrow \Sigma^*$  is said to be **computable** if there exists a Turing Machine  $F$ , which on input  $w \in \Sigma^*$  terminates with just  $f(w)$  written on its tape, for any  $w$ .
- **Definition:** A language  $A$  is **mapping reducible** to a language  $B$  (denoted  $A \leq_m B$ ) if and only if there exists a computable function  $f : \Sigma^* \rightarrow \Sigma^*$  such that for every  $w \in \Sigma^*$ ,  $w \in A \iff f(w) \in B$ .



# Mapping Reducibility

- **Definition:** A function  $f : \Sigma^* \rightarrow \Sigma^*$  is said to be **computable** if there exists a Turing Machine  $F$ , which on input  $w \in \Sigma^*$  terminates with just  $f(w)$  written on its tape, for any  $w$ .
- **Definition:** A language  $A$  is **mapping reducible** to a language  $B$  (denoted  $A \leq_m B$ ) if and only if there exists a computable function  $f : \Sigma^* \rightarrow \Sigma^*$  such that for every  $w \in \Sigma^*$ ,  $w \in A \iff f(w) \in B$ .



$f$  is called a mapping reduction from  $A$  to  $B$ .

$$A_{TM} \leq_m \overline{\text{EMPTY}}_{TM}$$

$$L(M_w) = \begin{cases} \{w\} & \text{if } M \text{ accepts } w \\ \emptyset & \text{if } M \text{ does not accept } w \end{cases}$$

$$A_{TM} \leq_m \overline{\text{EMPTY}}_{TM}$$

**Solution:** Given input  $\langle M, w \rangle$  for  $A_{TM}$ ,  $f(\langle M, w \rangle) = \langle M_w \rangle$  defined as follows:

$$L(M_w) = \begin{cases} \{w\} & \text{if } M \text{ accepts } w \\ \emptyset & \text{if } M \text{ does not accept } w \end{cases}$$

$$A_{TM} \leq_m \overline{\text{EMPTY}}_{TM}$$

**Solution:** Given input  $\langle M, w \rangle$  for  $A_{TM}$ ,  $f(\langle M, w \rangle) = \langle M_w \rangle$  defined as follows:

$M_w =$  "On input  $x \in \{0,1\}^*$

1. If  $x \neq w$ , then reject.
2. Simulate  $M$  on  $w$ .

$$L(M_w) = \begin{cases} \{w\} & \text{if } M \text{ accepts } w \\ \emptyset & \text{if } M \text{ does not accept } w \end{cases}$$

$$A_{TM} \leq_m \overline{\text{EMPTY}}_{TM}$$

**Solution:** Given input  $\langle M, w \rangle$  for  $A_{TM}$ ,  $f(\langle M, w \rangle) = \langle M_w \rangle$  defined as follows:

$M_w =$  "On input  $x \in \{0,1\}^*$

1. If  $x \neq w$ , then reject.
2. Simulate  $M$  on  $w$ .

**Claim:**  $f(\langle M, w \rangle) \in \overline{\text{EMPTY}}_{TM} \iff \langle M, w \rangle \in A_{TM}$ .

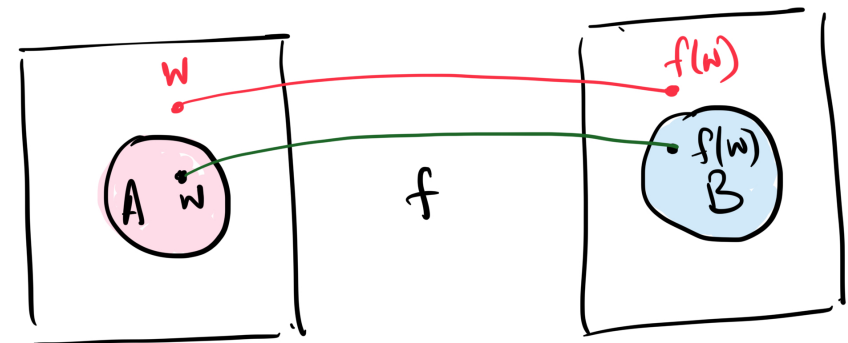
$$L(M_w) = \begin{cases} \{w\} & \text{if } M \text{ accepts } w \\ \emptyset & \text{if } M \text{ does not accept } w \end{cases}$$

# Mapping Reducibility: Properties

**Theorem:** If  $A \leq_m B$  and  $B$  is decidable, then  $A$  is decidable.

- **Proof:** Let  $M_B$  be a decider for  $B$ . We build a decider for  $M_A$  for  $A$  as follows:

$M_A$  accepts  $w \iff M_B$  accepts  $f(w)$   
 $\iff f(w) \in B$   
 $\iff w \in A.$



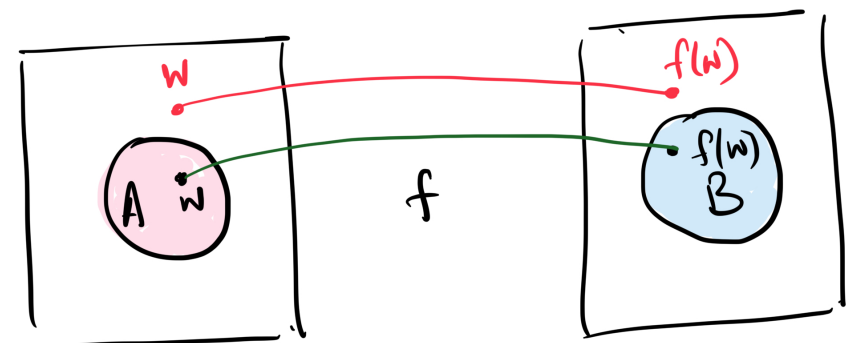
# Mapping Reducibility: Properties

**Theorem:** If  $A \leq_m B$  and  $B$  is decidable, then  $A$  is decidable.

“If  $B$  is at least as hard as  $A$ , and  $B$  is decidable then so is  $A$  .”

- **Proof:** Let  $M_B$  be a decider for  $B$  . We build a decider for  $M_A$  for  $A$  as follows:

$M_A$  accepts  $w \iff M_B$  accepts  $f(w)$   
 $\iff f(w) \in B$   
 $\iff w \in A .$



# Mapping Reducibility: Properties

**Theorem:** If  $A \leq_m B$  and  $B$  is decidable, then  $A$  is decidable.

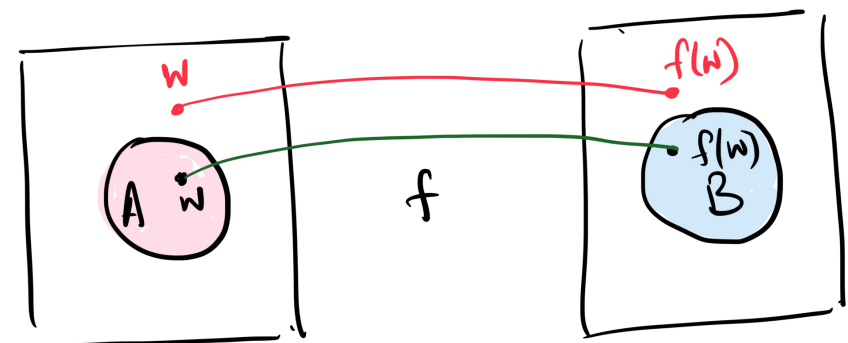
“If  $B$  is at least as hard as  $A$ , and  $B$  is decidable then so is  $A$  .”

- **Proof:** Let  $M_B$  be a decider for  $B$  . We build a decider for  $M_A$  for  $A$  as follows:

$M_A$  = “On input  $w \in \{0,1\}^*$

1. Compute  $f(w)$  .
2. Run  $M_B$  on  $f(w)$  and output its answer.

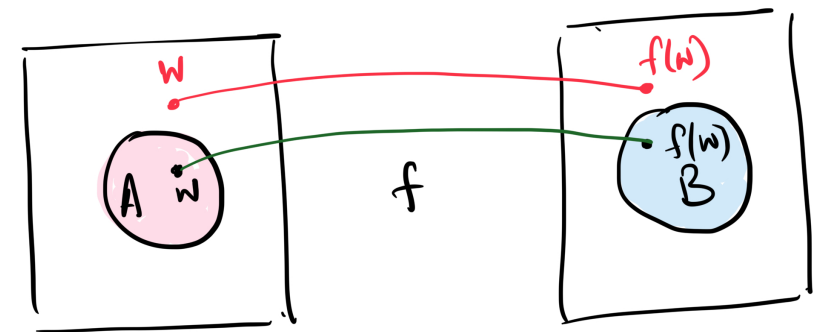
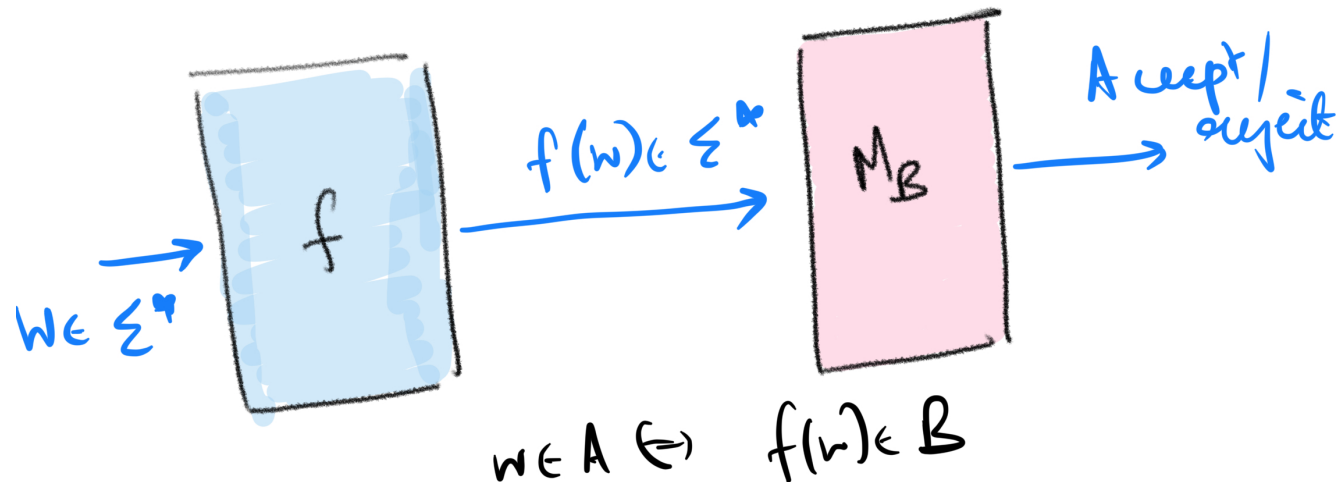
$M_A$  accepts  $w \iff M_B$  accepts  $f(w)$   
 $\iff f(w) \in B$   
 $\iff w \in A$  .



# Mapping Reducibility: Properties

**Theorem:** If  $A \leq_m B$  and  $B$  is decidable, then  $A$  is decidable.

A decider  $M_A$  for  $A$ :

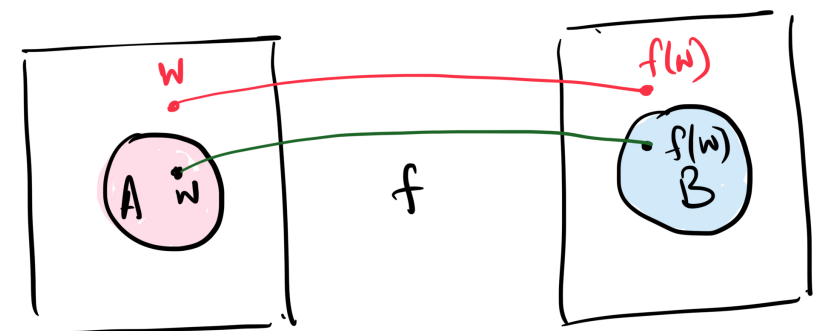
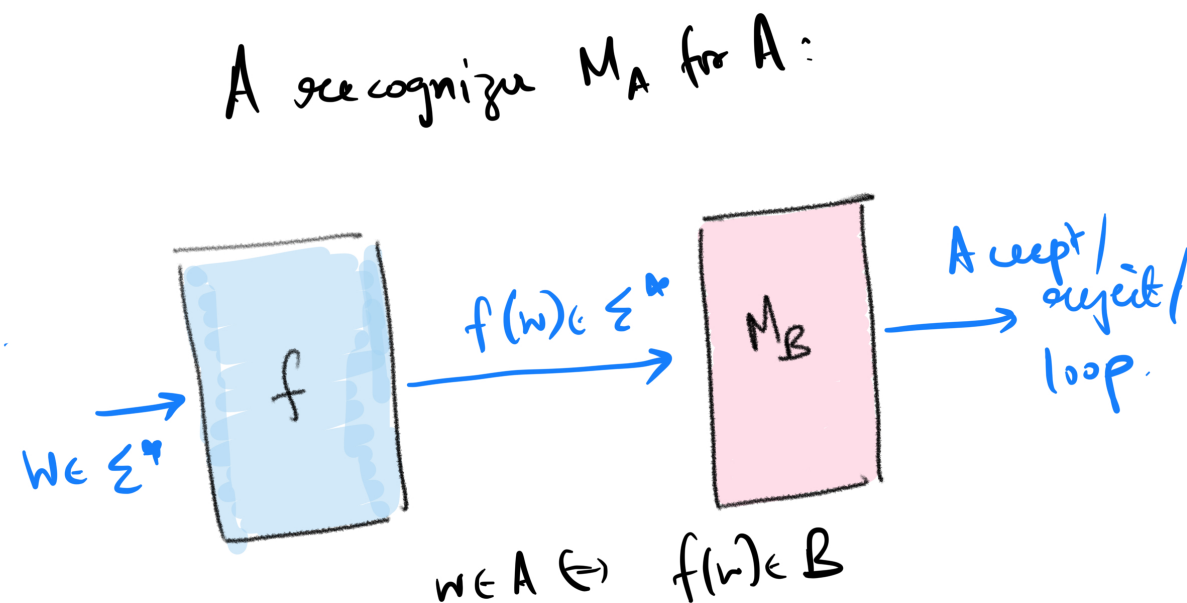


**Corollary:** If  $A \leq_m B$  and  $A$  is undecidable, then  $B$  is undecidable.

**A recipe for undecidability:** To show  $B$  is undecidable, find undecidable  $A$  and a mapping reduction from  $A$  to  $B$ .

# Mapping Reducibility: Properties

**Theorem:** If  $A \leq_m B$  and  $B$  is recognizable, then  $A$  is recognizable.

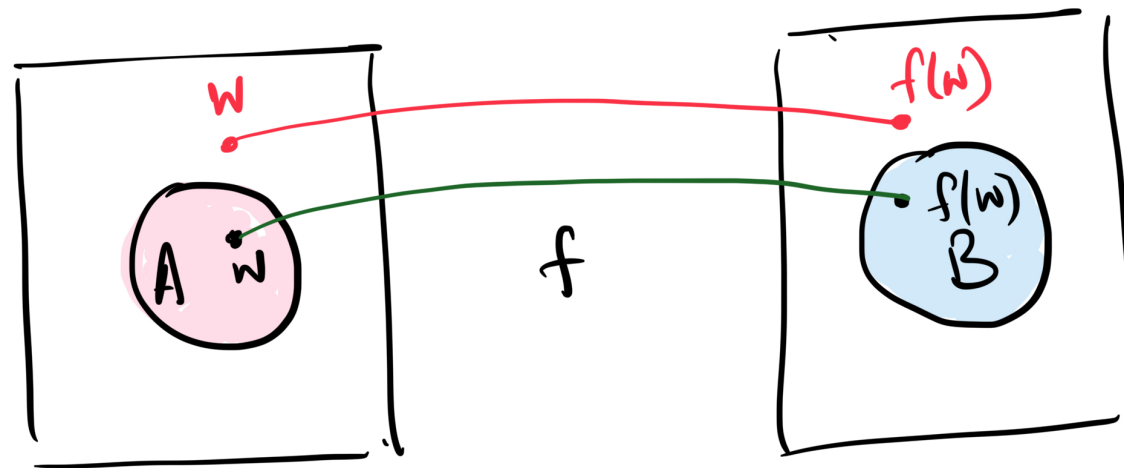


**Corollary:** If  $A \leq_m B$  and  $A$  is unrecognizable, then  $B$  is unrecognizable.

**A recipe for unrecognizability :** To show  $B$  is unrecognizable, find unrecognizable  $A$  and a mapping reduction from  $A$  to  $B$ .

# Mapping Reducibility: Properties

**Theorem:** If  $A \leq_m B$  then  $\bar{A} \leq_m \bar{B}$ .



# Using Mapping Reduction for Unrecognizability

# EMPTY<sub>TM</sub> is unrecognizable

- Let  $\text{EMPTY}_{TM} = \{ \langle M \rangle \mid L(M) = \emptyset \}$ .

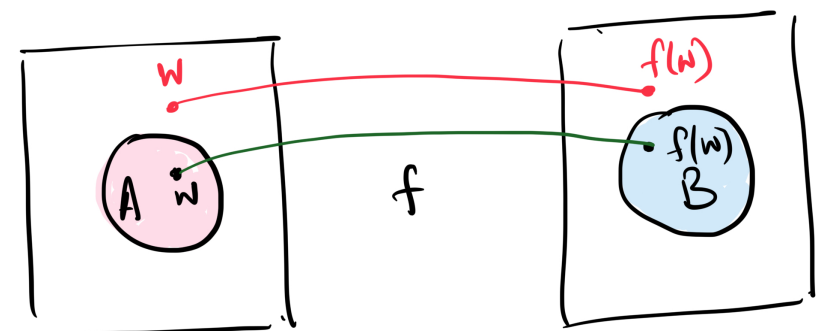
**Theorem:** EMPTY<sub>TM</sub> is undecidable.

- Proof:** Proof by contradiction. We will show  $\bar{A}_{TM} \leq_m \text{EMPTY}_{TM}$ .

**Reduction Function:**  $f(\langle M, w \rangle) = \langle M_w \rangle$

$$L(M_w) = \begin{cases} \{w\} & \text{if } M \text{ accepts } w \\ \emptyset & \text{if } M \text{ does not accept } w \end{cases}$$

$$\begin{aligned} f(\langle M, w \rangle) \in \text{EMPTY}_{TM} &\iff \langle M_w \rangle \in \text{EMPTY}_{TM} \\ &\iff L(M_w) = \emptyset \\ &\iff M \text{ does not accept } w \\ &\iff \langle M, w \rangle \in \bar{A}_{TM} \end{aligned}$$



# EMPTY<sub>TM</sub> is unrecognizable

- Let  $\text{EMPTY}_{TM} = \{ \langle M \rangle \mid L(M) = \emptyset \}$ .

**Theorem:** EMPTY<sub>TM</sub> is undecidable.

- Proof:** Proof by contradiction. We will show  $\bar{A}_{TM} \leq_m \text{EMPTY}_{TM}$ .

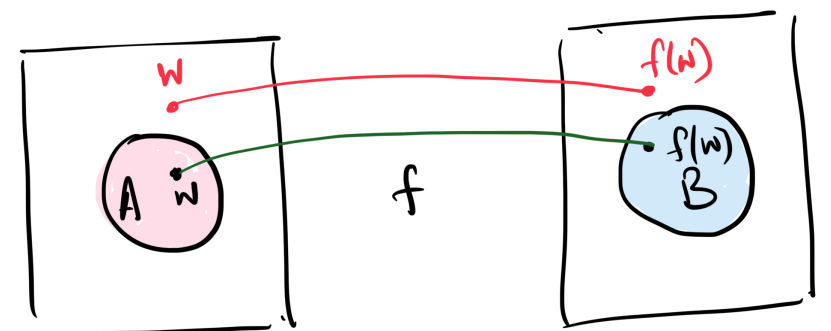
**Reduction Function:**  $f(\langle M, w \rangle) = \langle M_w \rangle$

$M_w =$  "On input  $x \in \{0,1\}^*$

1. If  $x \neq w$ , then reject.
2. Simulate  $M$  on  $w$ .

$$L(M_w) = \begin{cases} \{w\} & \text{if } M \text{ accepts } w \\ \emptyset & \text{if } M \text{ does not accept } w \end{cases}$$

$$\begin{aligned} f(\langle M, w \rangle) \in \text{EMPTY}_{TM} &\iff \langle M_w \rangle \in \text{EMPTY}_{TM} \\ &\iff L(M_w) = \emptyset \\ &\iff M \text{ does not accept } w \\ &\iff \langle M, w \rangle \in \bar{A}_{TM} \end{aligned}$$



# Reducibility Terminology

Why do we use the term “reduce”?

- When we reduce  $A$  to  $B$ , we show how to solve  $A$  by using  $B$ . We conclude that  $A$  is no harder than  $B$ . (This suggests the  $\leq_m$  notation).
- Possibility 1: We bring  $A$ 's difficulty down to  $B$ 's difficulty.
- Possibility 2: We bring  $B$ 's difficulty up to  $A$ 's difficulty.

# Summary

- Introduced the mapping reducibility to show undecidability/unrecognizability.
- Showed  $\text{EMPTY}_{TM}$  is undecidable and unrecognizable.
- Next time: more examples.